



Energy Systems Capacity Expansion Planning Under the Risk of Hurricane Damage

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Background

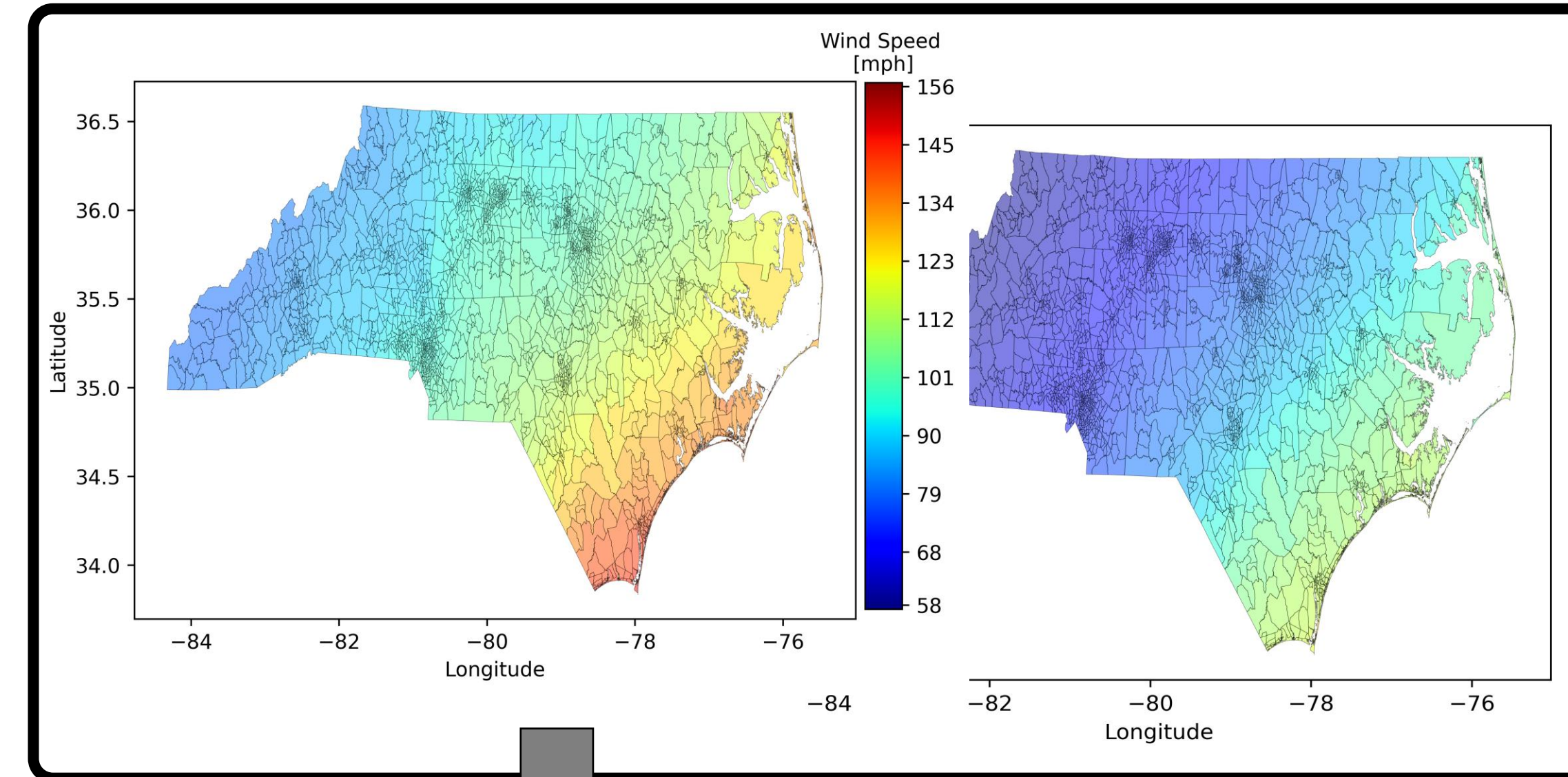
- ❖ **Hurricanes Can Lead To Significant Damage to Power Systems:**
 - ❖ The estimated cost to NC customers for the storm repairs made during 2018-2019 due to **Hurricanes Florence, and Dorian** account for more than **US\$1 billion** (Duke Energy 2021).
 - ❖ **Hurricane Irma** (2017) is estimated to cost **US\$1.3 billion** to Florida Power & Light customers (FPL 2018). **Hurricane Ian** (2022) US\$1.1 billion (NextEra 2022).
 - ❖ Capacity expansion models have been used to propose minimum cost paths for the evolution of the power sector. **How to incorporate the risk of hurricane damage into these models? When this representation is significant?**

Aims

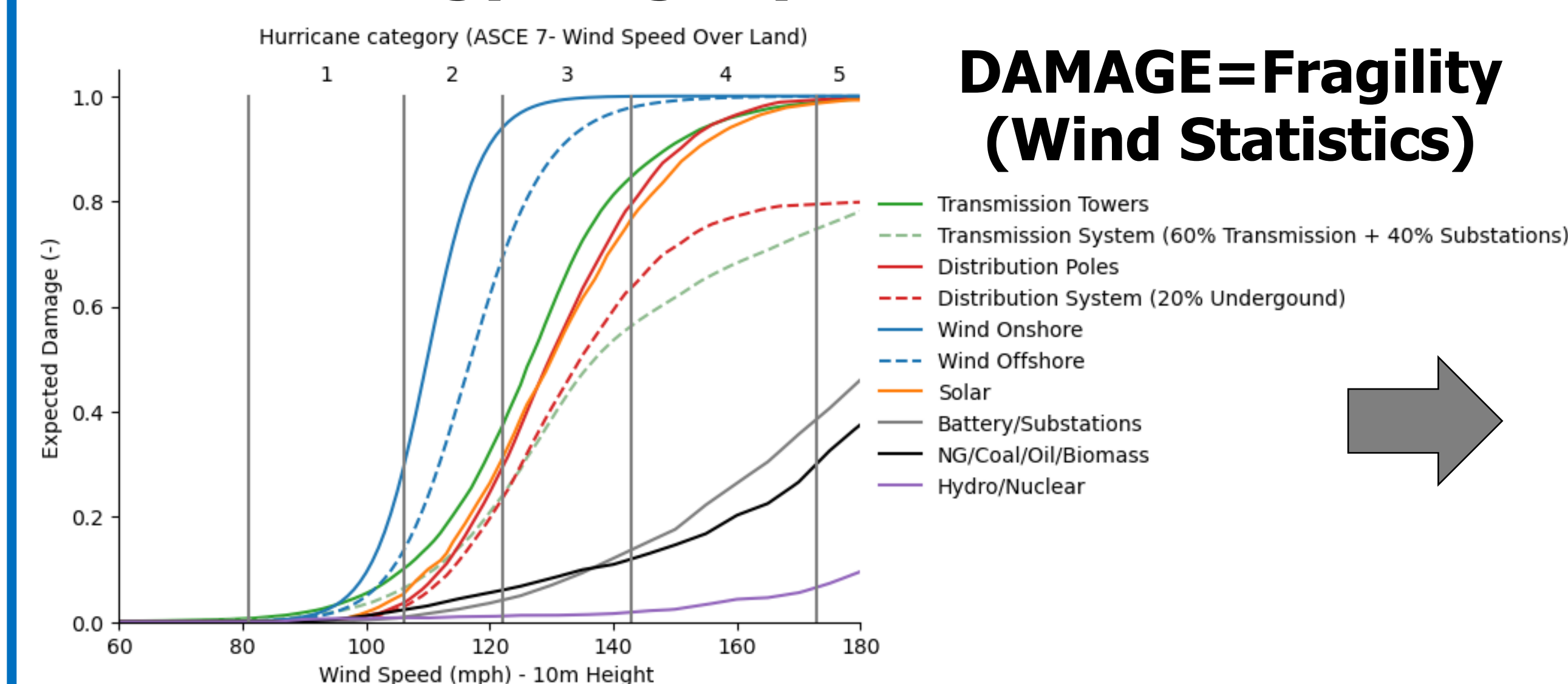
- ❖ Develop a capacity expansion model capable of representing the NC energy system and the risk of hurricane damage to different existing and future energy infrastructures.
- ❖ Use this model to assess the economic risks of hurricane damage and explore hedging strategies.

Modeling The Risk Of Hurricane Damage

Wind Statistics FEMA-HAZUS

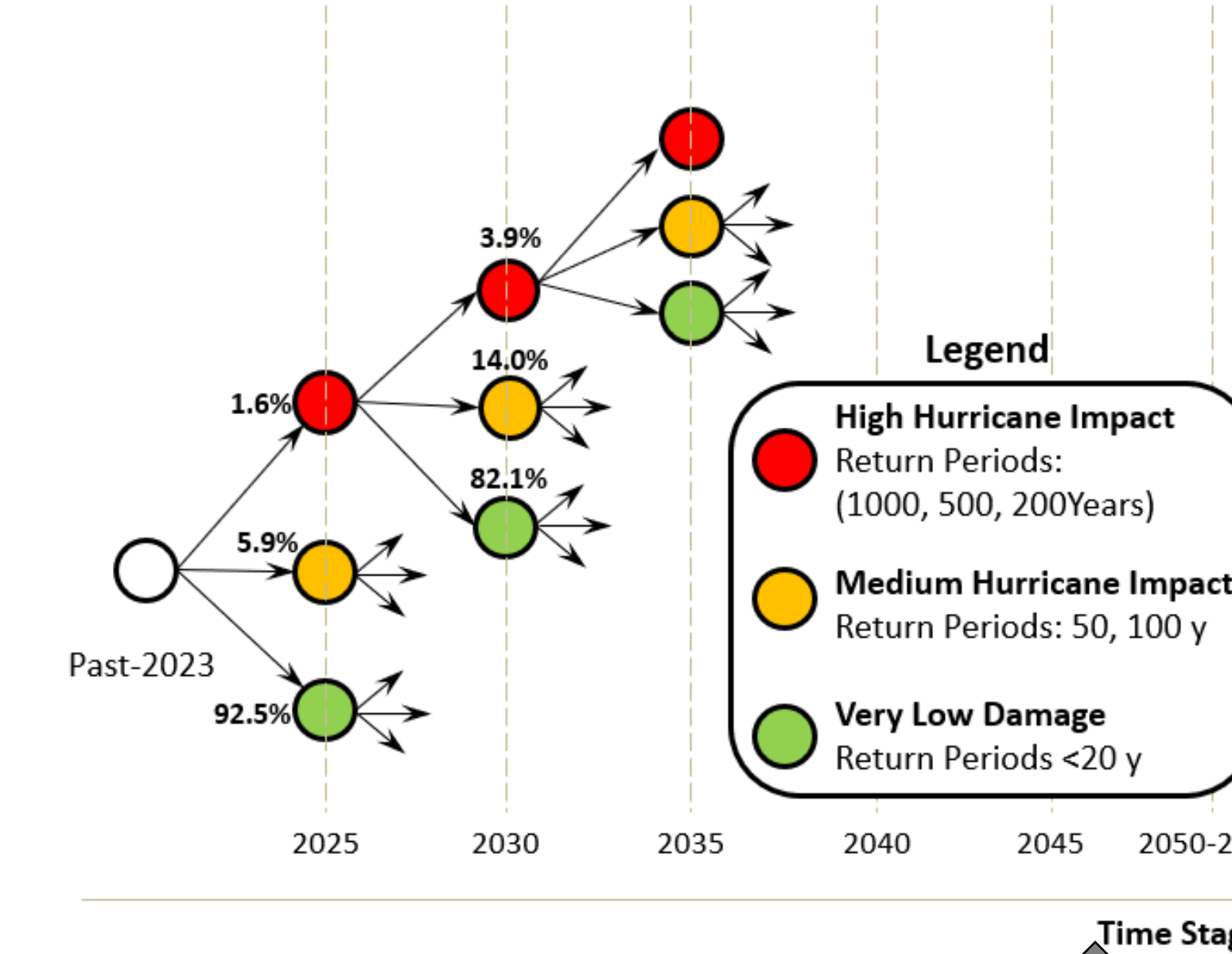


Technology Fragility Curves

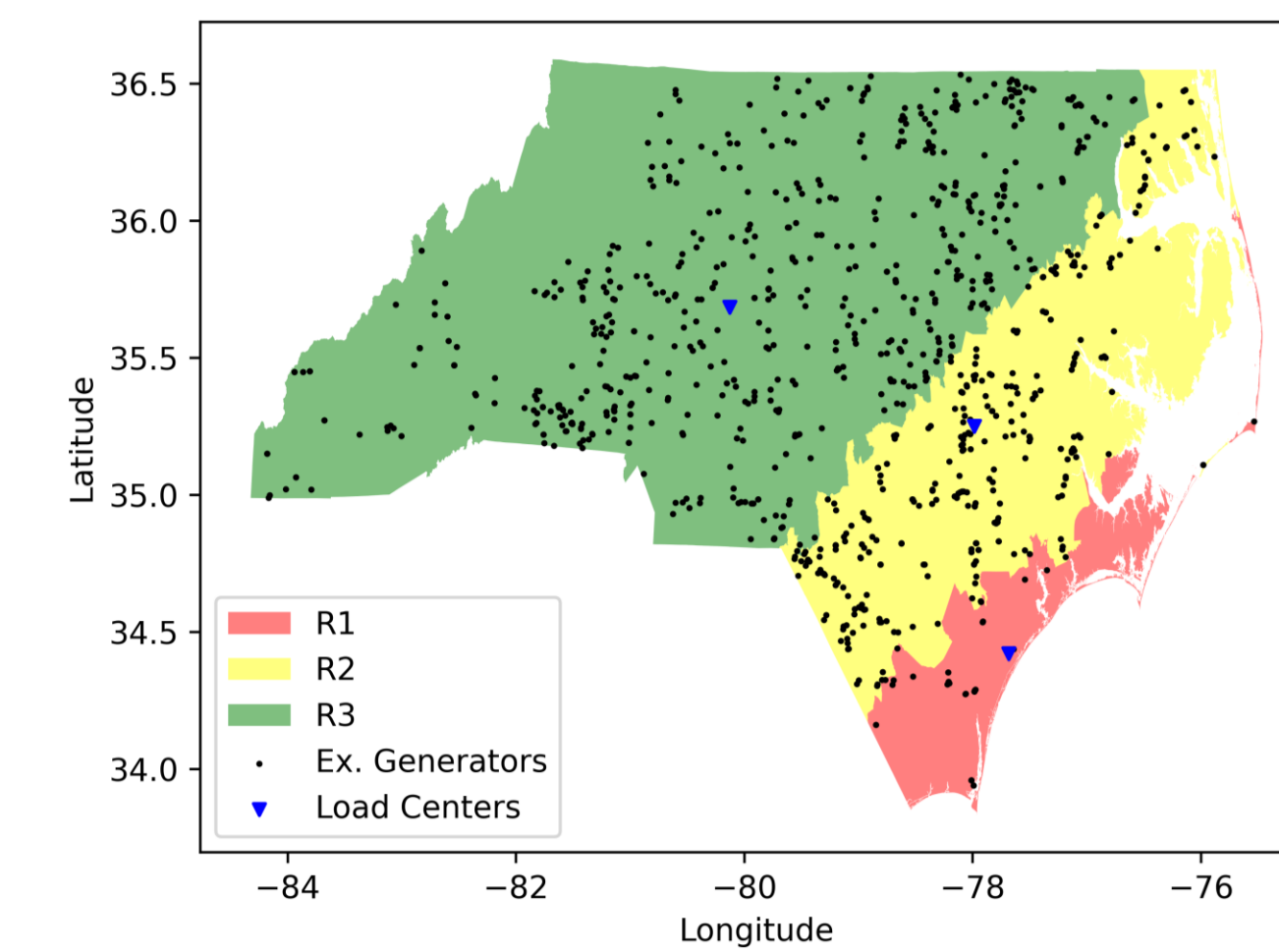


**DAMAGE= Fragility
(Wind Statistics)**

Scenario Tree



Discretize Regions Based on Risk



- Given different extreme wind scenarios with their corresponding probabilities, a stochastic optimization looks at each time step for the set of solutions (deployment size and type) that minimizes the **expected cost** to meet energy demand throughout the entire time horizon.

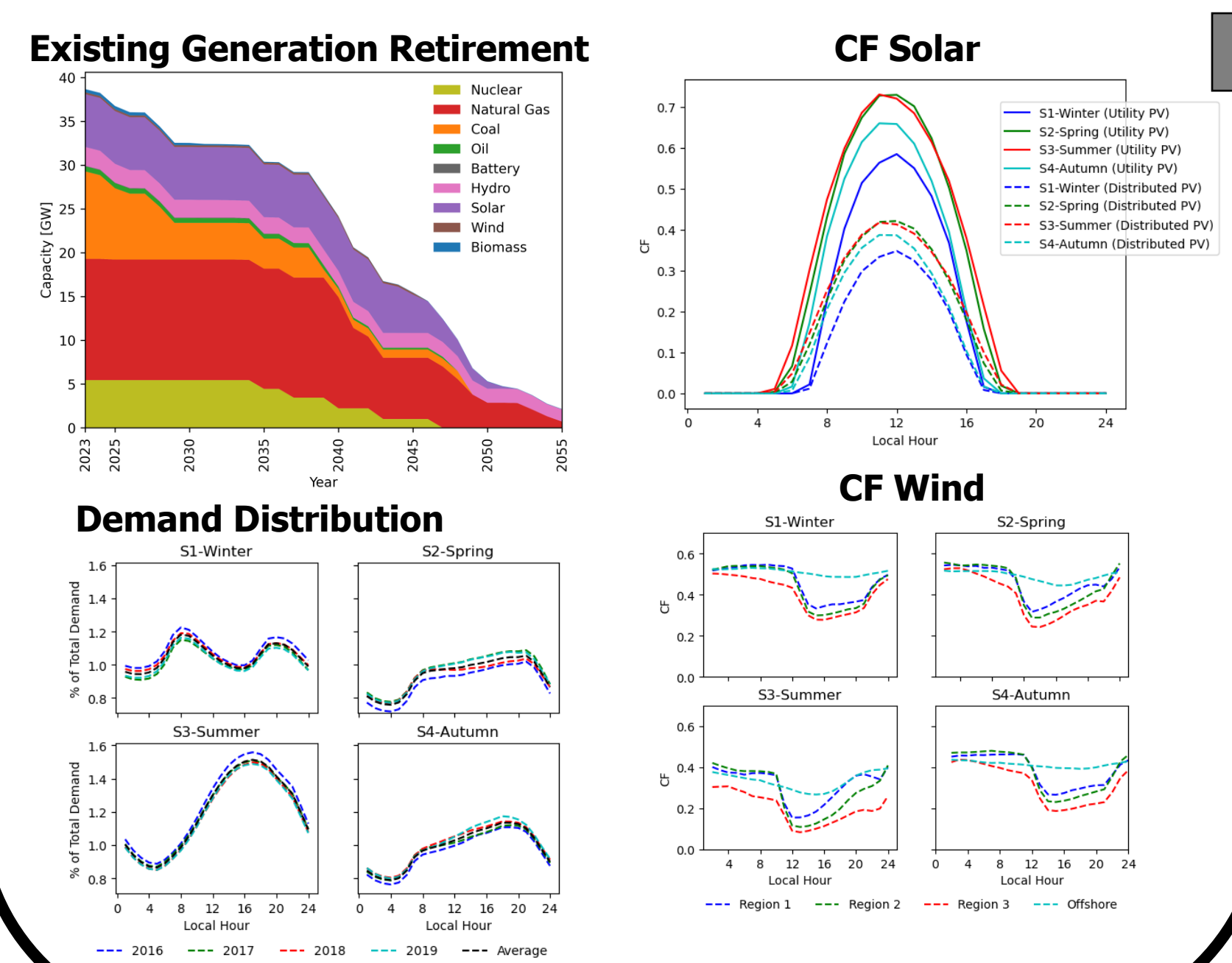
- In each scenario, damage to infrastructure is computed as the expected damage, conditioned on the hurricane intensities

- After the realization of a hurricane scenario, the existing capacity of any technology is reduced proportionally to its expected damage.

Capacity Expansion Model (Deterministic)

Data

- **The List of Parameters is Long:** Existing Generators Capacity; Emissions; Efficiencies; Fuel Prices; Technology Prices; Improvements in CFs; Max Capacity; Max Activity; Energy Imports; Energy Reserve Requirements, Capacity Credits; Technology Life; Government Policies/CO2 Targets, Demand Distribution...



Model:

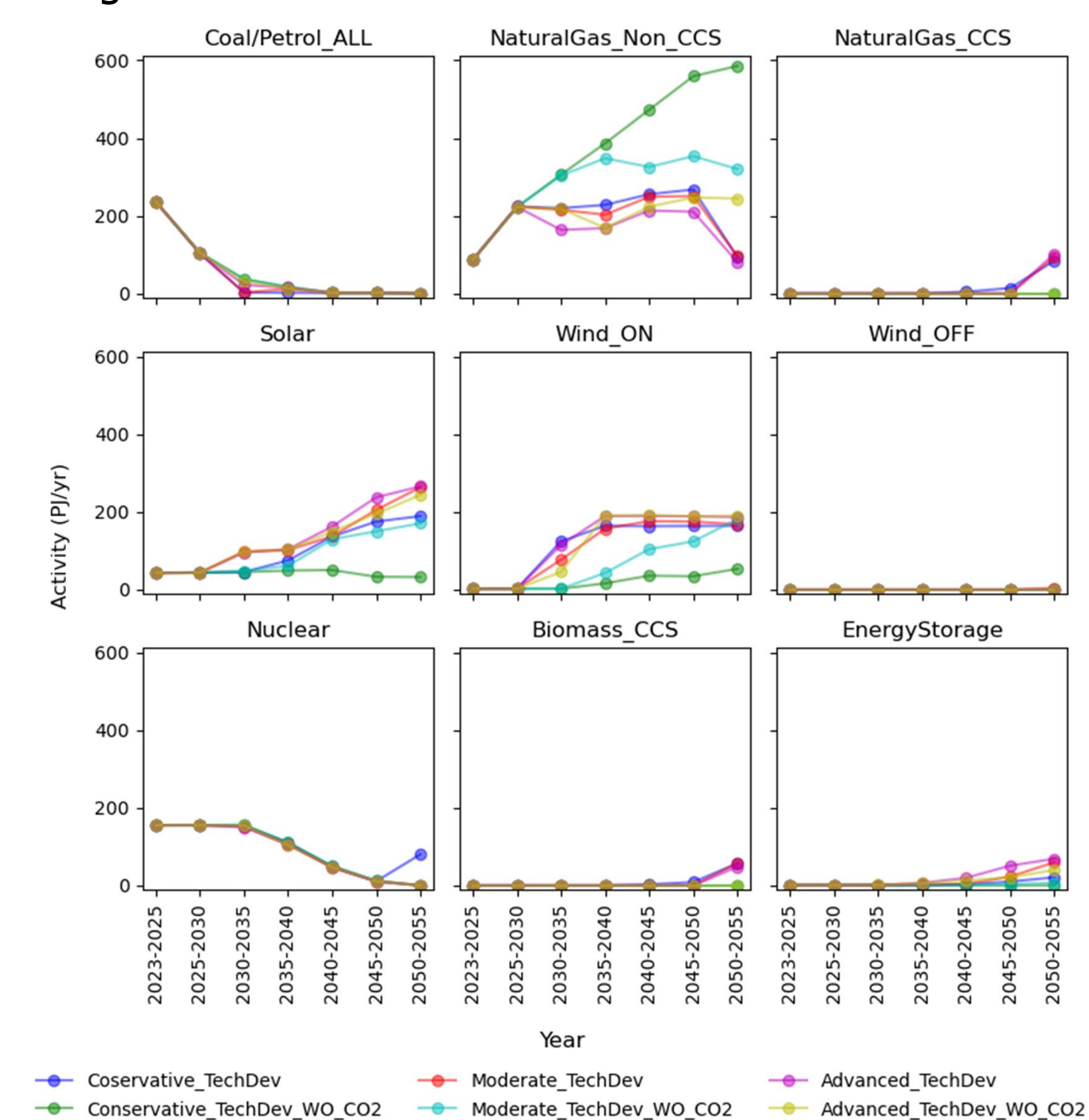
What is The Min Cost Solution to Meet Future Energy Demand?

Minimize: TotalCost = LoanCost + FixedCost + VariableCost	Objective Function
$ACT_{p,s,d,t,v,r} = \sum_{(i,o)} FO_{p,s,d,t,v,r,i,o} \quad \forall \{p,s,d,t,v,r\} \in \Theta_{act}$	Process activity
$(CF_{s,d,t,v,r} \cdot C2A_t \cdot SEG_{s,d} \cdot TLF_{p,t,v,r}) \cdot CAP_{t,v,r} \geq ACT_{p,s,d,t,v,r} \quad \forall \{p,s,d,t,v,r\} \in \Theta_{act}$	Technology Capacity
$\sum_{(i,t,v)} FO_{p,s,d,t,v,r,i,c} = DEM_{p,c,r} \cdot DSD_{s,d,c,r} \quad \forall \{p,s,d,r,c\} \in \Theta_{demand}$	Supply-demand
$FO_{p,s,d,t,v,r,i,o} = EFF_{t,v,i,o} \cdot FI_{p,s,d,t,v,r,i,o} \quad \forall \{p,s,d,t \in T - T^{st}, v,r,o\} \in \Theta_{flow}$	Process-Level Commodity Flow Constraint
$\sum_{(s,d,t,v,i,o)} (EAC_{t,v,r,i,o,e} \cdot FO_{p,s,d,t,v,r,i,o}) \leq ELM_{p,e,r} \quad \forall \{p,r,e\} \in \Theta_{demand}$	Emission limit constraint

Other Constraints and Bounds: Capacity/Activity Limits, Storage, Life of Technology, Retrofit Options, Planning Reserve Margin (Capacity Credit), Curtailment, Energy Imp/Exp

Preliminary Results

Total Activity for Different Technologies Under Conservative, Moderate, and Advanced Cost Assumptions from NERL-ATB. With and W/O CO2 Targets



Total Activity for Different Technologies Under Conservative Cost Assumptions, With CO2 Targets (Stoch Wind Speed +15%, Deterministic Model With No Hurricane)

